
From Prompts to Proof Obligations: Formal Sidecars as an Epistemic Interface for Trustworthy ML

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Abstract

Large language models make it increasingly cheap to attach formal, machine-checkable refinements to ordinary natural-language claims. This paper argues that such refinements should be treated not merely as technical aids, but as a new epistemic interface for trustworthy ML. The relevant artifact is no longer prose alone, but prose plus a formal sidecar: an explicit object specifying what proposition the answer commits to, what source or discourse scope it assumes, what evidence boundary it respects, what proof or checking obligation remains, and when the system must abstain. Using EG-VAR (Evidence-Grounded Verified Agentic Reasoning) as a case study, we show how empirical claims over tables (with the same pattern extending to typed APIs and structured sources) acquire exactly these five elements: a target proposition, a source/discourse scope, an evidence boundary, a checked proof obligation, and an explicit abstention condition. The philosophical point is not that natural language has a unique hidden logical form, or that proof assistants solve meaning. It is that cheap autoformalization turns interpretation into an auditable design object. Autoformalization therefore shifts part of the trustworthy-ML problem from model outputs to governed communicative artifacts.

1. Introduction: The Formalization Flywheel

Current debates over trustworthy machine learning concentrate on outputs, explanations, and statistical reliability (Grote et al., 2024): answers are correct or incorrect, explanations faithful or unfaithful, models calibrated or not. These diagnostics share a feature that gets little attention.

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They evaluate the prose the model produces, but they typically do not require the system to publish a checkable artifact specifying what the prose is *committed* to: which formal proposition the model claims to support, in which vocabulary, under which evidence boundary, through which proof obligation. Channel-level critiques of LLM output (Hicks et al., 2024; Frankfurt, 2005) correctly locate the defect here, since the model produces token sequences whose epistemic standing is opaque even to the model, but stop short of saying what would close the gap. Closing it requires changing what artifacts the system publishes, not what tokens it emits.

LLMs are quietly making such artifacts cheap. Autoformalization translates informal mathematics into Lean and Isabelle goals (Wu et al., 2022; Jiang et al., 2023), while proof-assistant-based theorem provers now solve olympiad-level and benchmark-scale formal mathematics (Google DeepMind, 2024; Xin et al., 2025). Lean matters for our case study because it combines a small proof-checking kernel with public LLM-facing theorem-proving infrastructure, reusable mathematical libraries, exact rational arithmetic for table quantities, and verifier feedback for search, repair, and training. As this technology moves beyond mathematics, an LLM can convert a claim over a table into a typed formal target (with the same pattern extending in principle to typed APIs and structured sources). Ordinary communication can then become *bimodal*: prose for coordination and persuasion, formal sidecars for checkable commitments, evidence boundaries, and inference. We call this trajectory the *formalization flywheel*.

This shift answers the workshop question directly. For LLM-generated empirical claims, trustworthiness should not be located primarily in fluency, confidence, or citation alone. It should attach to a published artifact: a natural-language answer plus a sidecar specifying its target proposition, source or discourse scope, evidence boundary, proof obligation, and abstention condition. Familiar worries about vagueness, context sensitivity, and discourse plurality then re-emerge as governance problems for that artifact. We use EG-VAR (Evidence-Grounded Verified Agentic Reasoning) in Section 4 as a concrete case study of this sidecar architecture on empirical claims over tables.

2. From Natural-Language Answers to Formal Sidecars

A *formal sidecar* for a natural-language claim or document is an LLM-assisted refinement consisting of: (i) a *target proposition* in a typed formal language; (ii) the *source or discourse scope* under which the target is meaningful; (iii) an *evidence boundary* identifying which observations may license the target; (iv) a *proof obligation* in a specified kernel or policy that, if discharged, certifies the target; and (v) an *abstention condition* specifying when the system must decline rather than guess. A sidecar is not a paraphrase. It is a separate artifact that travels with the prose and exposes what the prose is being asked to commit to. The proof obligation is the part most readily checkable by an automated procedure, but the sidecar’s philosophical content is broader: it makes the target proposition and the source scope first-class objects of audit. Crucially, sidecars do not certify truth simpliciter. A VERIFIED status means certified under the declared formal/evidential boundary; semantic faithfulness to English, source authority, and social legitimacy remain separate audit targets.

Several operations are often collapsed but have different failure modes. Claim formalization chooses the proposition. Source grounding connects it to attested observations through a per-source *lift*, an audited rule from source-ontology facts to world-ontology predicates. Proof search tries to find a derivation; proof checking validates a submitted proof against the trusted kernel. Authorization decides who may bind prose to a sidecar, and versioning tracks how the binding survives schema or institutional change. Mathematics autoformalization mostly studies formalization, search, and checking (Wu et al., 2022; Jiang et al., 2023); empirical claims add source grounding, authorization, and versioning because names, tables, API fields, and domain conventions require interpretive lifts.

The philosophically interesting object is not only the checked proof. It is the sidecar that tells us what the prose is being asked to commit to.

3. Autoformalization as an Epistemic Interface

3.1. From Answers to Sidecars

An ordinary LLM answer is hard to audit because the only available artifact is the prose. Autoformalization changes the artifact: the answer now carries a formal sidecar with the five elements introduced in §2.

In a closely related philosophical-logic proposal, Šekrst (2024) applies Artemov-style justification logic (Artemov, 2008) to LLM hallucinations, treating outputs as requiring explicit justification terms and source-strength annotations. The sidecar object is a complementary construction at a

different grain: it packages a justification together with the proposition being committed to, the source scope, the evidence boundary, and the proof obligation. Šekrst’s framework asks *how* an LLM output can be justified; the sidecar artifact additionally exposes *what* the output has committed to, *which* source semantics ground that commitment, and *which part* of the commitment a kernel has checked. Autoformalization, on this reading, turns an answer into a sidecar-bearing communicative object.

3.2. From Translation to Refinement

It is tempting to model the relationship between prose and sidecar as *translation*: the prose has a hidden logical form that the LLM uncovers. This is the wrong frame for two reasons.

First, the standard objections to one-to-one translation (Quine, 1969) block any reading that treats a sidecar as the unique logical content of the prose. Context sensitivity, vagueness, and discourse plurality are not accidents of natural language to be removed by a sufficiently careful formalizer; they are features that any candidate sidecar must take a stance on. Second, in practice multiple incompatible sidecars are often defensible: “top 5 by revenue” may be lifted as the five rows with largest revenue, or as the rows whose revenue rank is at most five, with different behavior on ties; “in September” may mean September of any year, or September of the year fixed by the surrounding context. Both readings are reasonable; neither is the prose’s hidden meaning.

We propose a different frame. A sidecar is a *refinement* of the prose: it stabilizes and narrows the commitments of the original utterance for specific purposes (auditing, composition, abstention, downstream inference). Sometimes the refinement is *conservative*: the sidecar makes explicit what competent participants already take the claim to require. Sometimes it is *interventionist*: it forces a choice of vocabulary, boundary, threshold, or source scope that had previously been left implicit. Neither is a defect; both must be visible. Refinement is an act of elicitation, not a discovery of an already-fixed meaning. Multiple candidate refinements can coexist, be compared, audited, and revised. Some sidecars may be empirical, some causal, some normative, some provenance-bearing. There need not be one canonical refinement.

The goal is not to discover the one true logical form of a sentence, but to make candidate refinements explicit enough to use, dispute, compose, and revise.

3.3. From Model Trust to Sidecar Governance

Once sidecars are the audit object, trust attaches to artifacts rather than to fluency: published vocabularies and source-

to-world bridges (made concrete as a source ontology (L1) and world ontology (L2) in §4), formal goals, proof objects, tool attestations, and typed output states such as VERIFIED or ABSTAIN. A source owner may authorize a lift, a model provider may publish a candidate formalization, and an auditor may accept or reject a proof object. Reliabilist accounts in epistemology, which range over belief-forming processes generally (Goldman, 1979), have recently been imported into ML (Grote et al., 2024) where reliability is often operationalized as statistical behavior of the model; the sidecar object adds an artifact-level component of warrant (formal safety and replayability of a declared commitment) alongside any process-level reliability claim.

Trustworthy ML systems should publish their sidecars, not merely their answers.

4. Case Study: EG-VAR

We illustrate the sidecar object on empirical claims, where the gap between informal prose and formal vocabulary is sharpest. Mathematical claims are closed under a formal library: their target vocabulary is fixed, and the autoformalizer’s job is to pick the right propositions inside that vocabulary. Empirical claims are open in a way that mathematics is not. They involve named entities, tables, API fields, and domain-specific conventions that no formal library knows about until a curator commits to a particular mapping. Bridging this gap requires an interpretive act we call a *source lift*: a step that turns a source-ontology fact into a world-ontology claim about an entity and a property. Whether the lift faithfully captures the source’s intent is not a property the kernel can check; it is exactly the kind of judgment a curator or domain expert must vouch for, and exactly the kind of judgment that a sidecar makes auditable rather than hidden.

EG-VAR (Evidence-Grounded Verified Agentic Reasoning) is organized around two typed ontologies and four runtime layers separated by trust level. The *source ontology* (L1) describes source-side facts (cell positions, column names, and values for tables; field paths and values for APIs) without committing to what those source facts mean in the world. The *world ontology* (L2) describes entities, properties, and relations, and is bounded and stable across sources. A *tool layer* executes storage queries against the source and returns *L1 attestations* (typed attestation records of the query and its result). A *per-source formalization layer* supplies a curator-audited *dictionary* (a mapping from the source’s columns into L2 entities and properties) and a menu of *lifts*, lemmas that promote L1 attestations into L2 world facts under the source’s interpretive commitment. A *Lean 4 kernel* is the only component permitted to label an L2 claim VERIFIED, the typed state signifying that the claim has descended from L1 attestations through audited lifts and a kernel-checked

proof; if no proof goes through, the state is ABSTAIN. Two untrusted LLM roles split the work: the *formalizer* proposes the L2 goal and cites audited lifts; the *solver* proposes tool calls and tactics. The kernel can certify VERIFIED only from attested tool records plus audited lifts; unaudited lifts are unavailable to proof (Appendix A expands). The published answer is extracted from the checked proof. Figure 1 sketches the pipeline.

The L1/L2 split is not an engineering convention dressed up. L1 names the formal ontology of the source: what counts as a row, cell, or field within a particular source interface. L2 names the formal ontology of the world-directed assertion: what counts as a state, rate, or comparison. The source lift is the audited interpretation boundary: neither raw data nor a generic law of logic, but a curator’s published commitment about when an L1 fact may stand as evidence for an L2 claim. EG-VAR treats that boundary as a co-equal trust artifact rather than hiding it inside an adapter.

Lean 4 is one natural kernel choice; the conceptual move is venue-agnostic. It is useful here because it provides a small trusted checker, exact rational arithmetic, equality/order reasoning, finite-data infrastructure for predicates such as `ArgmaxWhere`, `count`, and `sum`, and kernel feedback for LLM search and repair (Google DeepMind, 2024; Xin et al., 2025). Fine-tuning on that feedback is a natural extension, not a claim of this paper.

A small example fixes the intuition. For the toy claim “Alabama has the highest HIV incidence rate” against a stipulated source table *T* (versioned state-level rates, aggregate rows excluded), a minimal sidecar has the following shape:

Target prop.	<code>ArgmaxWhere</code> : state strictly maximizes <code>hivIncidence</code> under condition
Source scope	table <i>T</i> version; row domain; aggregate-row policy; uniqueness required (tie $\Rightarrow$ ABSTAIN)
Evidence bdy.	attested values for every row in domain; audited lift <code>hiv_incidence_rate $\rightarrow$ hivIncidence</code>
Proof oblig.	Alabama’s value $>$ every other in-domain row’s value
Abstention	dictionary fails to cover prose, ties at max., or proof fails $\Rightarrow$ ABSTAIN

The interpretive moves (which column counts as “the rate”, how ties are handled, and what rows are included) are visible in the sidecar rather than buried in the model’s output.

The pipeline separates fluent assertion, formal commitment, empirical grounding, proof checking, and abstention into distinct acts. In counterfactual stress tests of the kind reported in (Wu et al., 2024; Ming et al., 2025), where supplied evidence contradicts a model’s parametric prior (EG-VAR instantiates this case for tables), the typing discipline blocks one specific failure mode: the kernel does not accept VERIFIED for a target unless the proof descends from attested observations through an audited lift. Source-faithfulness in the broader sense additionally requires that the formalizer’s target and lift be the intended ones, which remains a sidecar

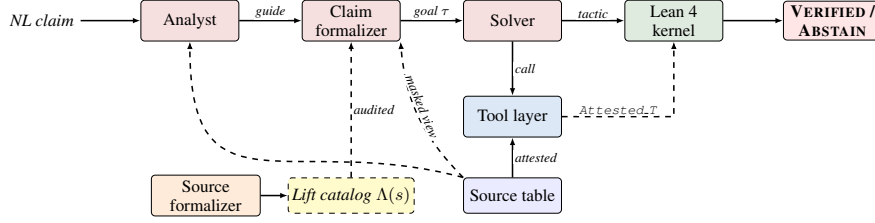


Figure 1. Multi-agent EG-VAR pipeline. Untrusted LLM roles propose goals, tactics, and tool calls; an offline curator publishes the audited lift catalog $\Lambda(s)$ for source s ; the runtime tool layer returns attested observations; the Lean 4 kernel accepts VERIFIED only when the proof descends from attested premises through audited lifts. Dashed arrows mark audited or attested boundaries.

audit question rather than a kernel guarantee. The point is mechanism, not benchmark scores.

The main residual risk is semantic: a wrong formal target can still be correctly proved. When sidecars are exposed for audit, it appears as an auditable error in the *formalizer or lift*, such as a wrong matching strategy, operator, column, condition, or composition structure, each an interpretive move an auditor can inspect or revise. Mistakes are still mistakes; the difference is that they appear at the sidecar layer rather than inside the model’s prose, turning the residual into a target for philosophical and linguistic intervention rather than a property only retraining can reduce.

Related work. The sidecar position sits at the intersection of autoformalization (Wu et al., 2022; Jiang et al., 2023), justification logic (Šekrst, 2024; Artemov, 2008), channel-level critiques of LLM hallucination (Hicks et al., 2024; Frankfurt, 2005), and reliabilist epistemology (Goldman, 1979; Grote et al., 2024), but shifts the central artifact to a typed object stating which proposition has been selected, which source semantics ground it, which evidence boundary applies, and which proof obligation has been checked (Appendix F expands).

5. Governance Questions for a Formalized Communication Layer

The hard philosophical questions begin *after* we take the flywheel seriously. A first agenda can be stated as candidate norms. *Disclosure*: VERIFIED answers should publish their sidecars. *Contestability*: sidecars should expose the formalization choice and source lift, not only the proof result. *Plurality*: divergent practices may require multiple scoped sidecars rather than one canonical logical form. *Non-upgrade*: plausible, unsupported, or merely cited claims should not be silently promoted to VERIFIED. *Authority*: source lifts need explicit provenance about who is entitled to bind source facts to world claims. These are not merely engineering requirements; they are norms for the public uptake of machine-generated commitments, specifying when a community is entitled to treat an output as evidence-bearing rather than merely fluent.

These norms raise composition, revision, and ambiguity questions. Locally valid clinical, insurance, and patient-facing ontologies may compose into false equivalences; evolving schemas may change the commitment rather than its surface representation; and law, diplomacy, or policy may require ambiguity to be preserved as alternatives rather than collapsed into one proof obligation. A concrete instance: `Diagnosed("MDD", patient, t)` can be lifted from claims-data billing records under a coding convention or from clinician notes under a clinical-reading convention. The two lifts can yield different VERIFIED sidecars for the same prose claim about a depression rate. *Authority* over which lift may certify a downstream use (actuarial pricing, public-health reporting, court testimony) is exactly what sidecar governance must adjudicate. The kernel checks the formal claim, not the intended English; sidecar audit is a precondition for, not a substitute for, audit of the interpretive layer.

Two readings should be ruled out. The position does not claim that proof assistants settle meaning: meaning lives in the source lift and dictionary, which are interpretive commitments published for audit, not theorems. Nor that the sidecar replaces prose. Channel-level diagnoses of LLM output (Hicks et al., 2024; Frankfurt, 2005) correctly locate the defect (fluent assertion severed from truth) but mislocate the fix: not better tokens, but a change to the artifact published alongside the prose. The sidecar is what reviewable practice can act on.

6. Conclusion: From Translation to Governance

LLM-mediated autoformalization is changing the medium of communication. The artifact that matters for trustworthy ML is the answer plus its formal sidecar. The central question shifts from whether natural language has a unique hidden logical form to how cheap formal refinements should be generated, authorized, composed, revised, and governed. The flywheel does not solve meaning; it makes meaning commitments cheap, explicit, and inspectable.

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A. EG-VAR Technical Sketch

EG-VAR is organized around two typed ontologies and four runtime layers. The *source ontology* (L1) describes a source as stored; the *world ontology* (L2) describes entities, properties, and relations. The *tool layer* executes attested storage queries against the source and returns L1 attestations (typed attestation records of the query and result). The *per-source formalization layer* is a curator-audited collection of dictionaries and lifts that connect L1 to L2 under the source’s interpretive commitment; this is the trust artifact, audited once per source. The *Lean 4 kernel* is the only component permitted to construct `Evidence Verified  $w$` for a world proposition w ; it does so via `mkVerified`, whose hypothesis requires an `Attested.T` term produced by the tool layer. The untrusted LLM roles are split: a *formalizer* proposes formal goals; a *solver* proposes tool calls and proof tactics. The kernel checks the solver’s proof against the submitted formal goal and attested premises; the formalizer’s target is published for audit, not semantically certified.

Sidecar contract. The formalizer is required to cite which audited lifts from the per-source lift menu it commits to; citing only audited lifts is part of the sidecar contract. Lifts not in the audited menu are unavailable to the proof and so cannot certify `VERIFIED`. This is what makes the source-lift boundary a published, contestable object rather than an internal implementation detail.

Why Lean 4. Lean is a mature target for LLM-assisted mathematics and theorem proving (Google DeepMind, 2024; Xin et al., 2025; Achim et al., 2025; Varambally et al., 2025). Its rational-arithmetic and decidable-equality libraries and tactics support the numerical comparisons that empirical table claims most often invoke; the kernel checks the resulting proof object. EG-VAR’s per-source libraries define table predicates such as `ArgmaxWhere`, `count`, and `sum` on top of this infrastructure, and the kernel-checkable proof object provides verifier feedback usable for formalizer search and repair.

A claim “which entity has property P ?” compiles to a dependent-pair goal in which P encodes the selection policy (e.g. `ArgmaxWhere`, `UniqueWhere`, or a domain-specific `Answer.Q`), so that a kernel-checked witness is the intended answer rather than an arbitrary one:

$$\Sigma (e : \text{String}) (v : \text{Value}), \text{Evidence Verified } (P \ e \ v).$$

The published answer is the first projection of the kernel-checked witness. If no witness is produced, the system returns `ABSTAIN`.

B. Safety Boundaries

EG-VAR’s safety properties are stated in plain English as follows.

No unsupported VERIFIED. Every source-observation premise in the dependency closure of a `VERIFIED` output is an attested tool result.

No unchecked deduction. Every accepted proof step type-checks in the Lean 4 kernel under the declared axiom set.

Out of scope. These properties do not claim semantic faithfulness of the natural-language-to-formal target, nor correctness of the trusted source schema and lift. These are *interpretive* matters and remain the sidecar’s audit surface; they are exactly where Section 5’s questions apply.

C. Running Example in Full

Claim: “Alabama has the highest HIV incidence rate.”

- **Source table** (schema): columns `state (string)`, `hiv_incidence_rate (rational)`.
- **Dictionary**: `state  $\mapsto$  Entity (State); hiv_incidence_rate  $\mapsto$  Property ("hivIncidence")`.
- **L2 vocabulary**: world-level `HasProperty(s, "hivIncidence", v)` (entity-property predicate; `argmax` sidecar invokes `ArgmaxWhere` which is defined over this predicate family).
- **Goal** (dependent pair):

$$\Sigma (s : \text{String}) (v : \text{Value}), \text{Evidence Verified } (\text{ArgmaxWhere "hivIncidence" Cond.all } s \ v).$$

- **Tool call:** `table_query` returning attested column values.
- **Proof:** kernel checks the Sigma’s witness $v = v_s$ (the attested `hivIncidence` value for row s) and $\forall s' \neq s. v_s > v_{s'}$ from the attested payload.
- **Result:** VERIFIED with witness $s = \text{"Alabama"}$ and replayable proof object.

D. Formalizer Disagreement Cases

A short catalogue of utterances with multiple defensible sidecars. Each case is interpretive, not syntactic; the disagreement is about which formal commitment best refines the prose, not about parsing.

- “*Top 5 by revenue.*” (A) Five rows with largest revenue, ties broken arbitrarily; (B) rows whose revenue rank is at most five, including all ties.
- “*Sales in Greece.*” (A) Rows with `country = "Greece"`; (B) under a binary Greece-or-Cyprus partition, `country  $\neq$  "Cyprus"`.
- “*In September.*” (A) Month = September across all years; (B) September of the year fixed by surrounding discourse.
- “*Total.*” (A) Sum over data rows excluding a pre-aggregated TOTAL row; (B) read the TOTAL row directly.
- “*The leading state.*” (A) Argmax over the metric in scope; (B) the state with the highest population in the table.

Each disagreement is the sort of move that becomes a typed audit record under autoformalization, rather than a hidden choice inside a free-text answer.

E. Relation to Philosophy and Linguistics

The position has anchors in several philosophical traditions and does not depend on adjudicating between them. Frege’s sense/reference (Frege, 1892), Kripke’s rigid designation (Kripke, 1980), Montague-style compositional semantics (Montague, 1973), and situation semantics (Barwise & Perry, 1983) all offer richer accounts of meaning than any single sidecar will encode. Quine’s indeterminacy (Quine, 1969) is the direct reason for treating sidecars as refinements rather than discoveries; inferentialist accounts (Brandom, 1994) locate semantic content in commitments, in the same family as the artifact-publishing stance of Section 3.3. We acknowledge the lineage without asking the position to settle it.

F. Relation to Technical Autoformalization Literature

The technical literature on autoformalization has focused on informal-mathematics-to-proof-assistant translation (Wu et al., 2022; Jiang et al., 2023; Lu et al., 2025) and LLM theorem proving in closed mathematical libraries (Google DeepMind, 2024; Achim et al., 2025; Xin et al., 2025; Varambally et al., 2025). Justification-logic accounts of hallucination (Šekrst, 2024; Artemov, 2008) share our central diagnostic that verified-grade claims must come with explicit warrant. The present paper differs in three respects. (i) It extends autoformalization beyond mathematics by making *source grounding* part of the sidecar; empirical claims need dictionaries and lifts that math claims do not. (ii) The philosophically interesting object is the sidecar, not the proof: the proof certifies the target, the sidecar specifies *which* target, in *which* vocabulary, under *which* source scope. (iii) Formalization is refinement rather than translation, which makes Section 5’s questions the natural sequel. Prior work asks how to improve autoformalization; this paper asks what kind of epistemic practice it creates. Closest technical systems use autoformalization or symbolic checkers to verify quantitative or logical reasoning; our concern is different: for empirical claims, the epistemic object is the sidecar that records the selected proposition, source semantics, evidence boundary, and abstention rule.

G. Evaluation Summary

A compact summary of EG-VAR’s empirical behavior; full implementation details and provenance will be released separately after de-anonymization.

- **Tier 1 unambiguous claims** ($n=120$, TableBench numerical reasoning): EG-VAR 120/120; same-tool same-model baseline 114/120 (95.0%); table-only single-shot 107/120 (89.2%).

- **Tier 1.5 counterfactual stress** ($n=20$ per panel, 5 domains $\times$ 2 frontier models, 2 injection regimes): *kernel-accepted* Verified outputs descend only from attested observations, so prior-leaning prose is blocked at the typing layer (observed in the panels; same-tool baselines score 87.8–100% on direct-injection panels and 60–83.3% on disambiguation panels, with the lower bounds being the failure cases). Semantic source-faithfulness of the formalizer’s target remains a sidecar audit question.
- **Tier 2 deployment-time formalization (Sonnet, $n=30$ v2 pilot)**: bucket-level breakdown is correct-as-gold (23), correct-after-disambiguation (4), kernel-supported sidecar with semantic-formalizer error (2; the sidecar’s target was formally proved but failed to faithfully refine the prose), honest abstain (1); rate of kernel-supported semantic error is $2/30 = 6.7\%$, exactly the typed formalizer-error surface the rest of this paper is about ($23+4+2+1 = 30$). These are not unsupported VERIFIED outputs; they are correctly typed proofs of the wrong target. The $n=120$ Level-1 sweep is documented separately.
- **Opus hard-case re-pass** (9 cases from a separate Sonnet batch, not the same 7 non-gold cases above): of 9 cases, 1 became correct-as-gold, 3 became correct-after-disambiguation, 1 was abstained by the formalizer after detecting an unsafe lift, 2 remain kernel-supported-but-semantically-wrong sidecars, and 2 are honest abstains the kernel surfaced rather than masking ($1+3+1+2+2 = 9$).

The Tier 2 residual is the philosophically informative number: it is not a hallucination rate but an audit of the interpretive layer, exactly the layer Section 5 asks philosophy and linguistics to help govern.

H. Glossary

Autoformalization. LLM-assisted formal refinement of an informal claim or document into a typed formal target.

Sidecar. A formal refinement that travels alongside prose (§2).

Kernel. A trusted proof checker; for EG-VAR, Lean 4 (de Moura & Ullrich, 2021). Only proof *checking* (not search) is part of the trusted base.

Formal safety vs. semantic faithfulness. *Formal safety*: the typing discipline prevents unsupported VERIFIED outputs and unchecked deductions (Appendix B). *Semantic faithfulness*: agreement between the formal target and the prose’s intended meaning; not a kernel property, audited at the sidecar layer.

Attestation / abstention. *Attestation*: a typed tool-execution record, precondition for kernel VERIFIED. *Abstention*: typed state when no proof is produced; distinct from “no answer”.